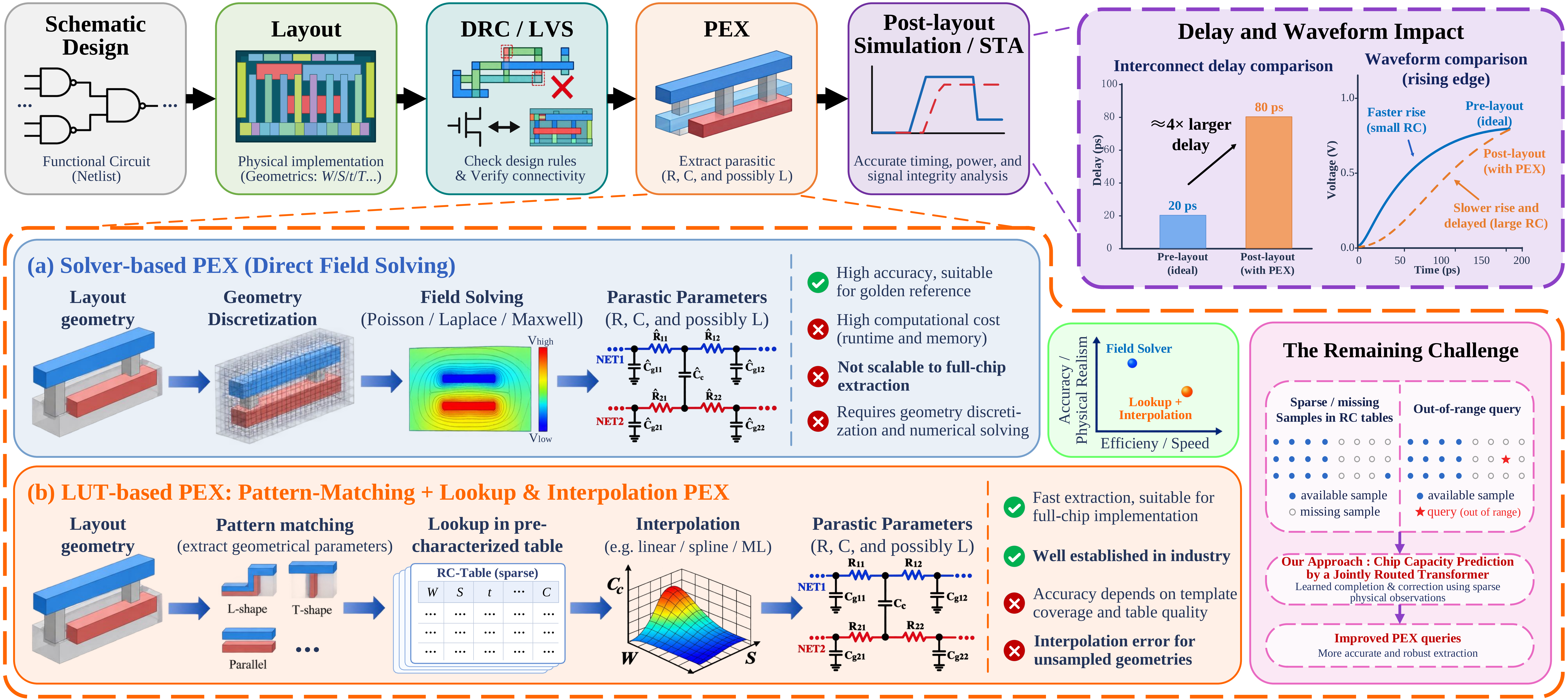
Schematic Design
Functional Circuit (Netlist)
Layout
Physical implementation (Geometrics: W/S/t/T...)
DRC / LVS
Check design rules & Verify connectivity
PEX
Extract parasitic (R, C, and possibly L)
Post-layout Simulation / STA
Accurate timing, power, and signal integrity analysis
Delay and Waveform Impact
Interconnect delay comparison
Delay (ps)
100
80
60
40
20
0
≈4× larger delay
20 ps
80 ps
Pre-layout (ideal)
Post-layout (with PEX)
Waveform comparison (rising edge)
Voltage (V)
1.0
0.5
0.0
0
50
100
150
200
Time (ps)
Faster rise (small RC)
Pre-layout (ideal)
Post-layout (with PEX)
Slower rise and delayed (large RC)
(a) Solver-based PEX (Direct Field Solving)
Layout geometry
Geometry Discretization
Field Solving (Poisson / Laplace / Maxwell)
Vhigh
Vlow
Parastic Parameters (R, C, and possibly L)
NET1
NET2
R̂11
R̂12
R̂21
R̂22
Ĉg11
Ĉg12
Ĉc
Ĉg21
Ĉg22
High accuracy, suitable for golden reference
High computational cost (runtime and memory)
Not scalable to full-chip extraction
Requires geometry discretization and numerical solving
Accuracy / Physical Realism
Efficieny / Speed
Field Solver
Lookup + Interpolation
The Remaining Challenge
Sparse / missing Samples in RC tables
Out-of-range query
● available sample
○ missing sample
● available sample
★ query (out of range)
Our Approach : Chip Capacity Prediction by a Jointly Routed Transformer
Learned completion & correction using sparse physical observations
Improved PEX queries
More accurate and robust extraction
(b) LUT-based PEX: Pattern-Matching + Lookup & Interpolation PEX
Layout geometry
Pattern matching (extract geometrical parameters)
L-shape
T-shape
Parallel
...
Lookup in pre-characterized table
RC-Table (sparse)
W | S | t | ... | C
Interpolation (e.g. linear / spline / ML)
Cc
W
S
Parastic Parameters (R, C, and possibly L)
NET1
NET2
R11
R12
R21
R22
Cg11
Cg12
Cc
Cg21
Cg22
Fast extraction, suitable for full-chip implementation
Well established in industry
Accuracy depends on template coverage and table quality
Interpolation error for unsampled geometries